

Sports intelligence - Lesson Plan

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Lesson Plan: QRU Sports Intelligence Lab(TM)

CHAPTER 1

Train Your Brain Like an Athlete Trains Their Body(TM)

1.1 Audience

General public: parents, caregivers, educators, coaches, and learners interested in using sports as a doorway into stronger understanding.

CHAPTER 2

Lesson Purpose

Every learner has the possibility to grow stronger in understanding. QRU Sports Intelligence Lab(TM) helps transform math learning into an athlete-training experience by connecting mathematics to sports, strategy, competition, practice, teamwork, decision-making, and reflection.

The learner becomes a Brain Athlete.

The goal is not only getting correct answers. The deeper goal is developing:

- Confidence
- Curiosity

- Persistence
 - Reasoning
 - Communication
 - Self-awareness
 - Problem solving
 - Lifelong learning ability
-

CHAPTER 3

Core Principle

"Children do not need more information.
They need stronger understanding systems."

CHAPTER 4

Big Idea

Athletes improve through:

- Repetition
- Feedback
- Adjustment
- Reflection

Learning follows a similar process.

In this lesson, mathematics becomes a mental sport. Learners practice thinking the way athletes practice their bodies.

CHAPTER 5

Memory Sentences(TM)

"Champions train their bodies.

Brain Athletes train their minds."

"Understanding creates confidence."

CHAPTER 6

Key Vocabulary

6.1 Multiplication

Multi = many

Plication = groups

Meaning: Many groups combined.

6.2 Intelligence

The ability to learn, understand, adapt, and solve problems.

CHAPTER 7

Lesson Objectives

By the end of this lesson, learners will be able to:

1. Explain multiplication as many groups combined.
 2. Connect sports practice to repeated thinking practice.
 3. Solve simple sports-based multiplication problems.
-

4. Explain their strategy instead of only giving an answer.
 5. Reflect on how practice, mistakes, and adjustment support learning.
 6. Recognize that struggling with math does not mean someone is "bad at math."
-

CHAPTER 8

Lesson Frame: The QRU Path

This lesson follows the QRU Translation(TM) learning sequence:

Question ? Understanding ? Practice ? Connection ? Application ?
Reflection ? Growth

Lesson Sequence

CHAPTER 9

1. Opening: The Brain Athlete Invitation

Begin with this idea:

Every athlete starts as a rookie. They enter practice, train, struggle, improve, and learn that champions are not created by talent alone. Champions are created through intentional practice.

Now connect that to learning:

A Brain Athlete trains the mind. The brain becomes stronger through repeated meaningful practice. Mistakes are not proof of

failure. In this lab, mistakes are training reps.

9.1 Teach the Analogy

- Brain = Muscle
 - Practice = Training
 - Mistakes = Training Reps
 - Knowledge = Equipment
 - Understanding = Game Strategy
 - Reflection = Game Film Review
-

CHAPTER 10

2. The Question

Ask:

10.1 Beginner Question

What is multiplication?

Guide learners toward the QRU definition:

Multiplication means many groups combined.

Example:

If a player completes the same number of actions again and again, multiplication helps us find the total.

CHAPTER 11

3. Understanding: Sports Make the Math Visible

Explain:

Math is not only something on a worksheet. Sports are full of patterns, groups, repeated actions, and decisions.

When athletes track performance, they often need to understand repeated groups.

11.1 Football Example

A quarterback completes 5 passes each quarter.

There are 4 quarters.

So:

$$5 \times 4 = 20 \text{ passes}$$

Meaning:

There are 4 groups of 5 passes.

11.2 Basketball Example

A player makes 8 free throws each practice.

The player practices 5 times.

So:

$$8 \times 5 = 40 \text{ free throws}$$

Meaning:

There are 5 groups of 8 free throws.

The math becomes connected to athletic progress.

CHAPTER 12

4. Practice: Train the Thinking

Invite learners to solve and explain.

12.1 Practice Problem 1: Football

A quarterback completes 5 passes each quarter for 4 quarters.

Ask:

1. What are the groups?
2. How many are in each group?
3. What multiplication sentence shows the total?
4. What is the answer?

Expected solution:

$$5 \times 4 = 20 \text{ passes}$$

12.2 Practice Problem 2: Basketball

A basketball player makes 8 free throws each practice for 5 practices.

Ask:

1. What is repeated?
2. How many practices are there?
3. What does the multiplication sentence mean?
4. What is the total?

Expected solution:

$$8 \times 5 = 40 \text{ free throws}$$

CHAPTER 13

5. Connection: Math Is Strategy, Not Just Speed

Teach clearly:

Multiplication is not:

- Memorizing without understanding
- Racing against others
- Measuring intelligence

Multiplication is a way to understand repeated groups and patterns.

A learner who needs more practice is not "bad at math." They may need:

- More practice
- Different explanations
- A different strategy

This preserves confidence and keeps the door open for growth.

CHAPTER 14

6. Application Activity: Design Your Own Sports Problem

Learners create their own sports-based multiplication problem.

They should include:

1. A sport or athletic activity
2. A repeated action
3. The number in each group
4. The number of groups
5. A multiplication sentence
6. The total
7. A short explanation of their strategy

14.1 Example Structure

A player does the same action several times across practices, quarters, rounds, or games.

Learners explain:

- "My groups are?"
 - "Each group has?"
 - "My multiplication sentence is?"
 - "The total is?"
 - "My strategy was?"
-

CHAPTER 15

7. Reflection: Game Film Review for the Brain

Reflection is like watching game film after practice. You review what happened, notice patterns, adjust your strategy, and come

back stronger next time.

Ask learners:

1. What did you understand better after practice?
 2. What felt difficult?
 3. What strategy helped you?
 4. What would you try differently next time?
 5. How can mistakes become training reps?
-

Coaching Guidance

The adult's role is to be a:

- Coach
- Facilitator
- Encourager
- Guide

The goal is not perfection.

The goal is helping the learner discover:

"I can learn difficult things."

Use language that builds capability:

- "Let's look for the pattern."
 - "What are the groups?"
 - "What is being repeated?"
 - "Explain your strategy."
 - "That mistake gives us information."
 - "Let's adjust and try again."
-

Common Misunderstanding

15.1 Myth

"I'm bad at math."

15.2 QRU Understanding

"You may need more practice, different explanations, or a different strategy."

Discussion Questions

CHAPTER 16

Beginner

What is multiplication?

CHAPTER 17

Intermediate

How can multiplication help athletes?

CHAPTER 18

Advanced

How does understanding patterns improve decisions?

Quiz Bank

CHAPTER 19

Recall Questions

1. What does multiplication mean in this lesson?
2. What is a Brain Athlete?
3. What does "multi" mean?
4. What does intelligence mean?

CHAPTER 20

Application Questions

1. A quarterback completes 5 passes each quarter for 4 quarters. What is the total?
2. A basketball player makes 8 free throws each practice for 5 practices. What is the total?
3. Create your own sports multiplication problem.

CHAPTER 21

Reasoning Questions

1. Why is multiplication more than memorizing facts?
2. How does sports practice compare to learning practice?
3. Why can mistakes be called training reps?

CHAPTER 22

Reflection Questions

1. What helped you understand the problem?
 2. What strategy did you use?
 3. What would you practice next?
 4. How did your confidence change as you worked?
-

Review Standards

When reviewing learner work, consider:

- Mathematical accuracy
- Age appropriateness
- Educational alignment
- Emotional safety

The review should support understanding, not shame. The purpose is to strengthen confidence, reasoning, and learning ability.

Closing Message

Sports can become a gateway to intelligence. A child does not need to believe they are "bad at math." They may simply need a better doorway into understanding.

QRU Sports Intelligence Lab(TM) teaches that every Brain Athlete can practice, reflect, adjust, and grow.

Every brain can train.

Continue your understanding at QRU:

